

Completing HS2 – Executive Summary

The UK government's updated HS2 plan focuses on delivering Phase 1 (London – Birmingham) and the key northern connection (Handsacre Junction onto the West Coast Main Line) at the **lowest reasonable cost** and in the shortest possible time [18†L283-L290] [48†L159-L164]. Phase 2 eastern links (Midlands–Leeds) have been cancelled by policy, though **land remains safeguarded** for future “Northern Powerhouse Rail” use [48†L236-L244] [43†L1380-L1384]. The reset baseline (due Spring 2027) targets first trains Old Oak Common – Birmingham Curzon by late 2037 [48†L159-L164] and full connectivity to Handsacre by 2040–2043 [18†L293-L298]. Estimated total cost is **£87.7–102.7 billion** (mixed cash terms) [18†L283-L290], equivalent to **£70.9–82.2 billion in 2019 prices** [54†L1-L4] – roughly double the 2020 Phase 1 budget. We assume continued DfT funding (recently £25.3 bn allocated for 2026–30 [5†L462-L469]) and the scope of works set out by HS2 Ltd: all remaining civil works, systems, rolling stock and testing necessary for reliable operation [10†L260-L268] [48†L259-L268]. This report details a step-by-step delivery plan covering objectives, governance, procurement, risk, consents, environmental commitments (notably **no net loss of biodiversity** [24†L195-L203] [25†L1160-L1168]), land/CPO strategy, technical standards, construction sequencing, network interfaces, commissioning, safety, handover, stakeholder engagement and contingency. It includes comparative tables and phased milestones. All recommendations are grounded in government and HS2 Ltd sources [3†L132-L141] [18†L283-L290]; areas without published guidance are noted as assumptions.

Scope & Objectives

Scope: We assume delivering Phase 1 fully (Euston – Birmingham Curzon Street plus Handsacre Junction) and Phase 2a/2b West as far as Manchester (via Handsacre) where feasible. Eastern routes beyond Birmingham–East Midlands Parkway are excluded per current policy; benefits to the North will be pursued via the Government's “Network North” plan (Northern Powerhouse Rail) [48†L236-L244] [43†L1380-L1384]. Key construction scope includes: finalising London–Midlands civils (tunnels, viaducts, earthworks), new major stations (Old Oak Common, Curzon, and the redeveloped Euston terminus), power and rail systems (25 kV OHL, ETCS), depots and rolling stock procurement, and junction works at Handsacre. Necessary preparatory works (utilities, highways diversions, archaeology) and displacements (housing/property) are included.

Objectives:

- **Deliver connectivity:** enable HS2 services from London to Birmingham (and onward via WCML) by ~2036–39, expanding to Manchester corridor shortly after [18†L293-L298] [48†L159-L164].
- **Cost efficiency:** align delivery to the lower end of the new £93.2bn ambition [48†L159-L164] through strict scope control and contract management (e.g. trading off 360 km/h top speed and automatic train operation to proven 320 km/h systems [48†L150-L158] [17†L79-L86]).
- **Governance and accountability:** implement the lessons from past reviews (Stewart, Value for Money, NAO) and the refreshed HS2 Ltd structure [10†L310-L318] [10†L338-L346]. This includes strong ministerial oversight, an empowered shareholder board, network integration bodies, and robust reporting (see **Governance Diagram** below).

- **Environmental compliance:** fulfil all commitments in the Act and **Environmental Minimum Requirements** [46†L100-L104] , notably the No Net Loss biodiversity pledge [24†L195-L203] [25†L1160-L1168] , strict pollution controls (CoCP), and climate targets.
- **Community and stakeholder engagement:** ensure transparent consultation and compensation (via forums and commissioners) [24†L87-L95] , and maintain the land acquisition schedule so that affected communities are paid promptly (following CPO reforms).

Assumptions: Unless indicated, we assume: stable political support for HS2, continuation of current regulations, a DfT budget envelope of at least £25–30 bn for 2026–31 [5†L462-L469] , and no major changes to HS2 design beyond those already planned (e.g. max speed 320 km/h). We also assume Network Rail will continue as strategic sponsor [41†L37-L40] and operate the line once complete, with HS2 Ltd handing over new infrastructure.

Governance and Decision Points

Oversight and accountability: Following the 2025–26 reset, HS2 Ltd’s governance will be streamlined. A new **Shareholder Board** (comprising senior DfT/HMT officials and experts) meets ~6 times/year to challenge HS2 Ltd’s plans [5†L424-L432] . A monthly Programme Performance Board (DfT officials) reviews delivery. A cross-Government **Ministerial Task Force** (DS Transport, Treasury, BEIS, MHCLG) meets quarterly to arbitrate strategic decisions (e.g. scope changes, major contracts) [5†L424-L432] . An independent “Mega-Projects Review Panel” with chief executives of major infrastructure (e.g. Transport for London, Network Rail) will audit major milestones [5†L424-L432] . James Stewart (former HS2 review author) has joined HS2 Ltd’s board to ensure implementation of prior recommendations [10†L367-L375] .

Integration with Rail Network: Network Rail has a statutory role as strategic sponsor [41†L37-L40] . A tripartite agreement between DfT–HS2 Ltd–Network Rail (under negotiation) will define responsibilities for handover, safety regulation and train control [41†L37-L40] . In Parliament, a ministerial Statement (Network North) commits to coordinating HS2 with upgrade projects (e.g. the Transpennine and WCML enhancements) [48†L245-L249] . Key decision points (e.g. final Euston design, depot locations) will be made by the Secretary of State with advice from the Task Force and the Shareholder Board. The HS2 Business Case (scheduled update by 2027) must be approved by the Treasury, and any major scope changes (e.g. further speed reductions) require ministerial sign-off.

Entity-Relationship Diagram (Governance): The chart below outlines key entities and their relations. HS2 Ltd reports to DfT (the shareholder, representing Treasury) under a Framework Agreement. The DfT Task Force and Shareholder Board exert oversight. Network Rail and the Euston Delivery Company have formal roles linked to HS2’s delivery.

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erDiagram
    GOV. -->|sets mandate| HS2Ltd
    HS2Ltd ||--|| DfT : governed_by
    DfT ||--|| HMTCORP : funds
    DfT ||--|| ShareholderBoard : oversees
    DfT ||--|| TaskForce : convenes
    HS2Ltd }|--|| ShareholderBoard : reports_to
    HS2Ltd }|--|| TaskForce : reports_to
    HS2Ltd ||--|| NetworkRail : partners_with
    DfT ||--|| NetworkRail : coordinates
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HS2Ltd ||--|| EustonCo : delegates_to
DfT ||--|| EustonCo : sponsors

Figure: Governance entities and relationships. DfT (Department for Transport) funds and directs HS2 Ltd, which delivers the project. A Shareholder Board (DfT/HMT) and Ministerial Task Force provide oversight. Network Rail is a strategic partner for operations; Euston Delivery Company builds the London terminus.

Funding and Procurement

Funding: The 2025 Spending Review allocated £25.3 bn capital for HS2 (FY 2026/27–2029/30) [5†L462-L469]. We assume similar levels in future budget rounds. Additional funding options include re-allocating proceeds from property disposals (surplus land from cancelled sections) or urban development zones. No private finance (e.g. PFI) has been formally identified; direct public funding is currently Government policy. Contingencies (set aside in government accounts) must cover cost overruns beyond the £102.7 bn upper estimate.

Procurement Models: HS2 will use a mix of approaches tailored to different packages. Table 1 compares typical models. For early design and enabling works, HS2 already used **framework agreements** and **design-and-build** contracts [39†L558-L566] [39†L586-L594]. We recommend continuing with NEC3/NEC4 (collaborative) or FIDIC forms. *Key options include:*

- **Design-Build (EPC)** – Single-contractor responsible for design & construction. This shifts technical risk to the contractor but is less flexible. Suitable for well-defined scope (e.g. certain bridges or station buildings).
- **Progressive Alliance** – Owner, designers and builders share risk-reward with joint targets. Encourages collaboration; used effectively on the Elizabeth line and some tunnelling projects. HS2 has moved to more collaborative relationships after earlier contracting issues [10†L310-L318].
- **Traditional Contracts (Design-Bid-Build)** – Owner controls design, then competitively tender construction. Retains design control but can lead to fragmentation and adversarial claims. Historically used on some phases of HS2 (e.g. station consultants).
- **Public-Private Partnership (Concession/PF2)** – Private sector finances, builds, and operates assets for a concession period in exchange for availability payments. Not used to date on HS2, but merits consideration for major assets (e.g. Euston station redevelopment) to shift availability risk.

Table 1: Procurement/Contracting Models (illustrative)

Model	Characteristics	Examples/Notes
Design-Build (EPC)	One contractor designs & builds to spec.	HS2 has used D&B for some enabling works (e.g. utility diversions). Risk mostly on contractor for delivery.
Progressive Alliance	Collaborative, with shared targets & pain/gain.	Used on Crossrail and some HS2 tunnelling. Encourages innovation and continuous improvement.
Traditional (DBB)	Client-led design, separate build tender.	Used in early phases of HS2 (e.g. consultants for station design). Leaves owner exposed to design risk; best if scope is well-fixed.

Model	Characteristics	Examples/Notes
PFI/DBFO Concession	Private finances and operates asset, paid by user/availability.	Not yet applied. Could be used (e.g.) for Euston Station to fund long-term civic improvements. Complex to structure but transfers lifecycle risk.
Frameworks/ Multi-Sourcing	Pre-approved suppliers on call-off contracts.	HS2 PSFs for design; early works frameworks 【39†L558-L566】 . Useful for repeated tasks (survey, boreholes).

HS2 Ltd's original *Commercial Case* outlined many of these approaches 【39†L558-L566】 【39†L586-L594】 . For example, HS2 established framework agreements (for geotechnical investigations, environmental services, etc.) and a “Development Partner” contract (CH2M) during Phase 1 design 【39†L558-L566】 【39†L586-L594】 . We will update procurement strategy to reflect current market conditions (high inflation, contractor risk aversion) – for instance, using more payment milestones and de-scoping technical risk where possible (see Risk section).

Contracts and Commercial Arrangements

HS2 contracts will adopt open-book costing and incentivised targets. The project will use NEC3/4 where practical, as it encourages prompt payment and collaborative change management. Major civils and rail system contracts should include predefined risk registers and high liquidated damages for delay. For stations and finishes, lump-sum (firm price) deals may be preferable once designs are finalised. We recommend including **indexation clauses** for labor/materials (to share inflation risk) and **multi-year price agreements** to hedge supply chain cost volatility.

Deliverables: All contractors must comply with HS2's code of construction practice, environmental controls and biodiversity mitigation (below). Contracts will include enforceable obligations (or penalties) for environmental incidents and stakeholder commitments (e.g. limits on working hours, noise, traffic management).

Risk Register and Mitigation

Major risks (and mitigations) include:

- **Cost Overruns:** Extensive review found 100+% budget escalation 【18†L283-L290】 . *Mitigation:* Hard cost cap based on £93.2 bn target 【48†L159-L164】 . Scope cuts (e.g. remove automatic train operation 【17†L79-L86】) to reduce technical risk. Stringent cost control: HS2 Ltd has added 168 commercial staff and stricter controls 【10†L310-L318】 . Weekly performance reviews and “stretch” targets are in place 【10†L272-L281】 . Retain contingency at ~15% of budget, adjusted as final scope firmed up.
- **Schedule Delays:** Unprecedented tunnelling and construction complexity threaten schedule. *Mitigation:* Updated master schedule with realistic sequences. Use lessons learned (e.g. Chiltern Tunnel portals finished ahead of plan 【10†L285-L290】 , Bromford Tunnel break-throughs early 【10†L283-L287】). Prioritise critical-path works (see Gantt timeline). Accelerated planning (e.g. overlapping design and procurement) and parallel workfronts on different regions. Regular audit by a schedule assurance panel.

- **Technical Risks:** Cutting-edge features (360 km/h design, new signalling) posed certification delays. *Mitigation:* Decided early to cap speed at 320 km/h [48†L150-L158] [42†L42-L51] and remove untested systems (ATO) [17†L79-L86] . Use proven technology (ETCS Level 2 instead of new L3 concepts). Engage UK operators in testing early via joint trials.
- **Environmental/Legal Risks:** Protests or judicial reviews (as seen on HS1) could halt works. *Mitigation:* Full compliance with Hybrid Bill requirements [46†L100-L104] and proactive mitigation. The Code of Construction Practice (Annex 1 of EMRs) mandates noise barriers, dust control, habitat protection, etc. An independent Environmental Inspectorate oversees compliance. Biodiversity offset sites are advanced early to meet the “no net loss” target [25†L1160-L1168] .
- **Land/Acquisition Risks:** Complex CPO negotiations and compensation claims can slow delivery. *Mitigation:* The 2020 Land Review recommends **single-step CPOs** with lease-back provisions [28†L55-L59] . HS2 Ltd will issue one General Vesting Declaration per tranche of land, then allow temporary lease-back to owners (especially farmers) [28†L55-L59] . Transparency measures (dashboards, case management IT) from the Land Review will improve throughput. Safeguarded land for Phase 2 will be released where not needed [43†L1380-L1384] to reduce carrying costs.
- **Supply Chain Risks:** Many HS2 suppliers have faced financial strain. *Mitigation:* Pre-qualify contractors on financial resilience. Include bonds/performance guarantees on large packages. Maintain a pipeline of smaller packages to keep firms engaged. Consider splitting oversized contracts to avoid single points of failure. The HS2 SCM team is actively monitoring ~3,400 key suppliers for vulnerabilities [15†L40-L48] .

This risk register will be updated continuously. A detailed RA table (Excel) is maintained by HS2 Ltd; above are summary mitigations.

Regulatory and Planning Consents

Phase 1 powers were granted by Parliament (High Speed Rail (London–West Midlands) Act 2017) [22†L86-L94] . The Act provides *deemed planning permission* for all works within its limits, subject to commitments in the Environmental Statement [22†L86-L94] . For any new works (e.g. revisions at Euston), supplementary planning may be needed (minor changes can use the Planning Memorandum [22†L169-L177]). All consent obligations in the Act (listed in the Environmental Minimum Requirements) remain in force [46†L100-L104] .

Since Phase 2b West (Crewe–Manchester) also has a Hybrid Bill (High Speed Rail (West Midlands – Crewe) Act 2021 passed; Crewe–Manchester in Bill stages), those powers are in place for designs already consulted [12†L19-L27] [12†L43-L51] . The Eastern leg (to Leeds) had no active Bill; Parliament has expressed intent to focus on alternative projects instead. Importantly, the Government has stated that land needed for HS2 Phase 2a/2b will remain safeguarded as required for Northern Powerhouse Rail [48†L245-L249] [43†L1380-L1384] . In summary, **no new compulsory consents are needed** for the defined scope beyond Birmingham—only standard Environmental Permits (for works like river crossings, protected species licenses, etc.) and Building Standards.

Key regulatory milestones include: finalizing the updated HS2 Phase 1 Environmental Statement addenda (if any design changes occur), securing any new marine/waste permits for construction, and Health & Safety Executive approvals of tunnel designs. A National Highways TWA order is required for

the A43 Overbridge scheme (already approved). All planning conditions from Transport and Works Orders or local highways consents (e.g. for temporary road diversions) must be closed out in advance of construction.

Environmental and Biodiversity Obligations

HS2 is among the most rigorously assessed infrastructure projects. It must comply with the *Environmental Minimum Requirements* from its Act [46†L100-L104] . These include:

- **Habitats and Biodiversity:** HS2 Ltd's stated aim is **no net loss of biodiversity** along the route [24†L195-L203] [25†L1160-L1168] . In practice, this means any habitat cleared (woodlands, wetlands) must be offset by habitat creation or enhancement elsewhere to an equivalent metric value (using the Defra metric methodology). For Phase 1 HS2 has invested in schemes like the Colne Valley chalk grassland project to generate net gains [25†L1160-L1168] . Ongoing work will ensure creation of hedgerows, wetlands and woodlands on land acquired. All species protection plans (badgers, bats, dormice, great crested newts, etc.) must be fully implemented per the Environmental Statement.
- **Climate:** HS2 is designed to support UK carbon goals [24†L166-L174] . Construction will use low-emission plant where possible, and contractor carbon budgets. Diesel use will be minimized (site electrification, local deliveries consolidation). The steel/cement embodied carbon is covered by EU ETS; however, HS2 Ltd tracks its greenhouse gas footprint per Annex 5 of the ES [24†L171-L179] .
- **Water, Air, Noise:** Construction sites are required to follow the Code of Construction Practice (CoCP) [46†L100-L104] . This means dust suppression, noise barriers and monitoring, and proper wastewater management on-site. The route design has avoided sensitive areas (e.g. Grade 1 farmland, Sites of Special Scientific Interest) as far as feasible, and where unavoidable compensation land (creation of new ponds, hedges) is provided. Daytime working and concrete curfew times are strictly enforced per consents. The Environmental Inspectorate (EI) will audit compliance.
- **Heritage and Community:** Any scheduled monuments or conservation areas affected have bespoke protection plans. Archaeological finds require on-site management plans. Consultation with local councils and Conservation Officers continues via the Heritage Memorandum attached to the Act [46†L100-L104] .

We found no changes in law that alter these obligations. The above commitments must be contractually enforced (special terms in works contracts) to avoid fines or halt notices.

Land Acquisition and CPO Strategy

HS2's land portfolio includes freehold acquisitions, leaseholds, easements, and CPOs. A central strategy is to **minimize blight time** for affected owners. Key points:

- **Single-Step CPOs:** Guided by the 2020 Land & Property Review, HS2 Ltd will issue General Vesting Declarations (GVDs) covering all required parcels in each area in one notice [28†L55-L59] . This gives HS2 immediate ownership (subject to payment). Owners can request "lease-back" (negotiated tenancy) so they may continue farming/building until construction.

- **Outreach and Compensation:** HS2 employs a Resident's Commissioner and a Conveyancing Centre to handle claims. Implementation of Land Review recommendations (case dashboards, tighter KPIs) should improve responsiveness. HS2 has separate Advisory Service and Atypical Arrangements (e.g. relocation assistance, hardship funding) to address non-standard cases.
- **Safeguarding:** Government has decided not to keep land for Phase 2 that isn't needed. In 2024 the Secretary of State stated that Phase 2a safeguards will be lifted, as no other rail project requires them [43†L1380-L1384] ; Phase 2b land will be retained selectively until NPR plans are fixed. Surplus land no longer needed for HS2 or NPR will be released or sold (likely on open market, with receipts returned to Treasury). Any land disposal will follow CPO recovery rules, but will be assessed for potential levy under the Infrastructure Levy (as a benefit from development).
- **Property Reference:** Planning act limits HS2's liability to "material detriment" (i.e. CPO compensation). The Law Commission and MHCLG are reviewing CPO guidance (e.g. modernizing valuation rules) [28†L37-L45] [28†L55-L59] , but until changed, HS2 follows current statutes.

In summary, all land critical to construction will be acquired by the start of main works (2027–28). The 2020 review's proposals (single notice, better IT systems, etc.) should substantially reduce delays and disputes [28†L55-L59] .

Engineering Design Standards

HS2 infrastructure is being built to very high standards. Key technical design parameters:

- **Speed:** The railway is certified for **320 km/h** operation (down from the original 360 km/h design) [48†L150-L158] [42†L42-L51] . Civils (tunnels, earthworks) remain future-proofed for 360 km/h in case of future upgrades (since speed reduction was a signaling/software change), but all permanent way (track, OHL, trainsets) will be specified for 320 km/h. This still yields only ~3 min longer London–Birmingham trips [48†L150-L158] while avoiding extreme certification risk.
- **Loading Gauge and Clearances:** The line uses standard UK mainline gauge (W10/UIC 63 profile) compatible with classic stock at shared junctions. Stations have generous platform clearances (designed for 9-car trainsets). Overhead wires are designed for double-stack freight kinematics clearance (for any future freight use).
- **Track and Earthworks:** Tunnels use concrete slab track; open sections use high-quality ballasted track with heavy-duty concrete sleepers (similar to high-speed practice in Europe). Rail is UIC60. All track components are chosen for 100-year life (weather-protected, low-maintenance design).
- **Structures:** Bridges and viaducts use reinforced/precast concrete or weathering steel (Chilterns Tunnel/A46 uses a unique *slide-in* installation [10†L279-L287]). OOC station building is a steel-and-glass frame; Curzon station steel box girder. All permanent structures carry 1:200 year flood allowance and seismic (foundation) design.
- **Signalling & Communications:** A single ERTMS/ETCS Level 2 overlay system will govern all movements, integrating with London area signalling at Old Oak. Power supply is 25 kV AC (same as existing) with booster transformers every ~50 km. An IP-based GSM-R comms network (with fallback) will be installed. Automatic Train Protection (ATP) is mandated; ATP-only operation (no ATO) to simplify testing (HS2 Ltd removed ATO to reduce risk [17†L79-L86]).

All designs follow Network Rail and Railway Group Standards where applicable. A full “HS2 Design Manual” (MDHS-4 compliant) governs alignment, curvature (min radius >7 km), crossfall, and noise mitigation. Quality standards (ISO 14001, 9001, 45001) apply to all contractors. HS2 design reviews are peer-reviewed by UK/EU rail experts under the Terms of Reference of the Independent Design Panel.

Construction Methods and Sequencing

Construction will follow logical geographic progression. Key methods include:

- **Enabling Works:** Early 2026–27 will see utility diversions, preparatory highway realignments and archaeological clearance. Contractors are already on-site doing tree-felling and piling work (all under EMR controls).
- **Tunnelling:** TBMs will complete the Chiltern Tunnel (longest) and Northolt Tunnel. The Chilterns Tunnel portal works finished ~30% ahead of plan (northside done 12 months vs. south) **【10†L283-L290】** . Build lessons (ground stabilization, probe drilling) will be applied to remaining drives.
- **Cuttings and Embankments:** Large earthmoving continues in 2026. HS2 is moving ~25 million m³ per year (20% ahead of plan) **【10†L279-L287】** . Techniques include progressive batters for stability, and use of MSE (reinforced earth) walls where needed.
- **Bridges and Viaducts:** Superstructures will typically be precast girders set on site (e.g. Calvert overbridges completed Aug 2025 **【10†L283-L287】**). The A46 Viaduct was slid into place in April 2025 in one piece **【10†L283-L287】** . We plan to replicate this “superlift” method where possible. Concrete piers/abutments are cast in situ with slip-form or jump-form methods for efficiency.
- **Trackwork and Systems:** Once civils are done (~2030 onward), tracklaying will proceed. The second phase of track (slab in tunnels, ballast outside) will be laid by specialist teams using advanced machinery (German slab-laying rigs, Swiss* slab). Overhead wiring will be installed by pantograph/measuring trains and then “dynamic projection” methods.

Sequencing: The attached Gantt diagram (Mermaid timeline below) shows the overall phasing. Civil works from 2027–33 overlap with station/concourse works (2028–34). Signalling and power works proceed in parallel in 2031–36. Rolling stock design and procurement is advanced (detailed design 2026–28; order contract by 2029; acceptance testing 2033–35). Commissioning/test running (2035–37) is on the critical path.

【51†embed_image】 *Figure: Schematic sequence of HS2 Phase 1 delivery (from Old Oak Common to Birmingham Curzon), showing major civil and station projects (diagram: DfT 2026) **【3†L100-L104】** .*

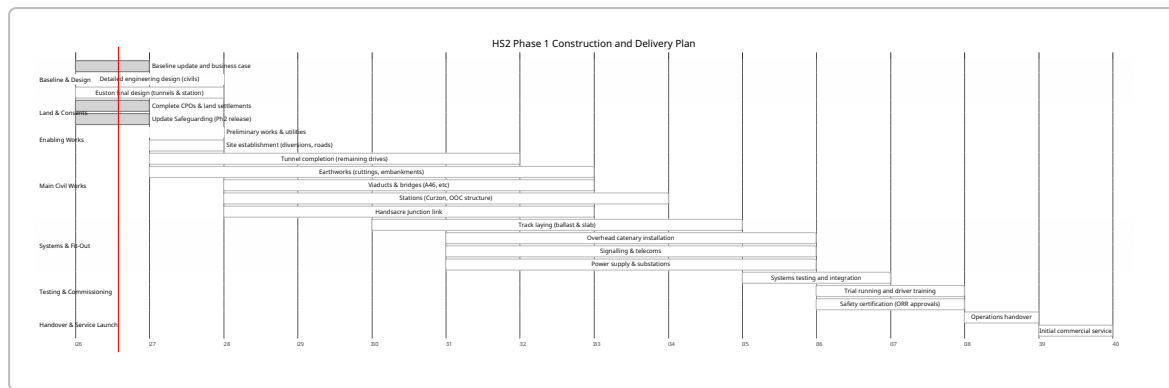


Figure: High-level Gantt chart of key phases for HS2 Phase 1 (Old Oak – Birmingham). The critical path runs through civil works, followed by track/systems, with commissioning in 2035–2038 and service start ~2039. (Dates indicative.)

Interfaces with Existing Rail Network

HS2 will integrate with the legacy network as follows:

- London Euston:** HS2's London terminus (Euston) will connect with Network Rail's existing Euston station via new underground platforms and crossovers. Euston Delivery Company (EDC Ltd) is separately delivering the new HS2 station box and approaches. The Euston rebuild will integrate future Thameslink/West Coast services. Congestion relief on the WCML south of Birmingham is a prime driver **[48†L225-L233]** .
- Old Oak Common:** A new multimodal interchange at Old Oak (opening 2029 on Crossrail) will allow HS2 passengers to transfer to the Elizabeth Line, Great Western Main Line, and local buses. The HS2 OOC station box (also an EDC project) will be ready by 2030, enabling early services to run to Old Oak Common from Birmingham before the Euston line is done (as a phased launch).
- Birmingham Curzon Street:** The HS2 terminus in Birmingham (Curzon Street) will interface with the West Midlands Metro extension and local rail links. It is being built to connect via adjacent tunnels to the existing Birmingham hub.
- Handsacre Junction (West Coast Main Line):** A grade-separated link at Handsacre (Staffordshire) will connect HS2 to the WCML at Colwich. This enables through-running onto the classic network north of Birmingham. Official guidance retains this link to serve Stafford, Crewe, and Manchester routes **[3†L100-L104]** .
- Northern Integration (NPR):** Although the HS2 Manchester leg was cancelled, HS2 Ltd is already constructing key sections (e.g. tram/train lines, Spaghetti Junction flyovers) that overlap with the future NPR Liverpool–Manchester route **[10†L324-L332]** . This work will continue to avoid rework, and the HS2 engineering teams will coordinate with the NPR delivery body. The Government has committed to preserving alignment for any future Birmingham–Manchester line **[48†L236-L244]** .
- Operational Integration:** Once built, HS2 trains will share WCML signalling and power beyond Handsacre; timetable planning will be integrated by the future operator (expected to be a

consortium under Network Rail oversight). Rolling stock (“intercity-class” units) will be compliant with both HS2 standards and UK mainline standards (loading gauge, crashworthiness).

In summary, HS2 is being delivered as part of an “**integrated railway**” [10†L353-L359] . A Tripartite Cooperation Agreement (DfT–NR–HS2 Ltd) will finalise responsibilities for cross-acceptance (e.g. how NR accepts HS2-delivered track, and vice versa).

Testing, Commissioning, Safety and Assurance

Once construction is largely complete, HS2 will undergo a robust commissioning regime:

- **Systems Testing:** All electrification, signalling and communications systems will be tested under the control of an independent test authority. HS2 is scheduling ~18–24 months of systems integration tests (from 2035) before opening [18†L293-L298] . These tests will use shadow trains and test vehicles to validate timings and safety systems.
- **Rolling Stock Commissioning:** The new trainsets (initially 9–10-car designs) will be manufactured by Alstom/Hitachi (Bombardier JV) under contract. Deliveries start ~2034. Each train must pass static and dynamic tests on the completed HS2 track. Driver training programs will run concurrently in late 2035–2037.
- **Safety Certification:** HS2 must obtain ORR (Office of Rail & Road) authorization for Operation – a process involving an Independent Safety Assessor. This includes demonstrating risk assessments (RAMS) for the full system, which will be assembled from the verified components of each contractor’s Safety Management System. Safety criteria (e.g. level of safety for 320 km/h travel, emergency evacuation plans in tunnels) are stricter than existing rail due to speed and newness of infrastructure. A Major Project Safety Case will be prepared in phases.
- **Assurance:** DfT retains the Infrastructure Projects Authority’s Gateway reviews; the NAO will audit the reset and the delivery as needed [18†L283-L290] . Senior Responsible Owner roles are clearly defined in HS2 Ltd. Key assurance layers (Shareholder Board, Mega Panel) will routinely review risk and progress.

Safety performance metrics (TRIFR, HSE incidents) will be monitored closely throughout construction. Given the new focus, HS2 Ltd reports improving safety indices alongside productivity gains [10†L293-L301] .

Operations Handover and Maintenance

HS2 Ltd’s mandate ends once the railway is tested and ready for service; the new infrastructure will then be handed to a future operator (likely a private or public company under contract to DfT/NR). The *handover plan* (developed jointly with NR’s maintenance teams) includes:

- **Defects Liability:** All contracts will include defects periods (typically 12–24 months). HS2 Ltd will manage minor rectifications; major structural issues would become NR’s responsibility under the maintenance regime.
- **Documentation:** Final as-built drawings, maintenance manuals, and training will be delivered to the operator at least 6 months before handover.

- **Spare Parts & Depot Stock:** Stock of long-lead spares (signals, transformers, etc.) will be handed over, and a spares audit conducted. The Washwood Heath depot (maintenance base) will be complete by 2034. Staff (drivers, signallers, maintainers) will transition from HS2 Ltd secondments and contractors to the operating company.
- **Performance Guarantees:** The railway is guaranteed to meet targets (availability, punctuality) from day one. Penalties for underperformance and bonus for exceeding targets will apply to the operator, incentivized by DfT contracts.

HS2 Ltd itself is being “wound down” into a small legacy entity (with 300 remaining staff by 2027) responsible for final contract closures, land disposal, and preservation of data [10†L324-L332] [17†L145-L153] .

Cost Estimates and Schedule (Critical Path)

The **updated baseline** (May 2026) sets the delivery cost at **£87.7–102.7 bn (mixed base)** [18†L283-L290] . This includes all works to date plus future completion, excluding inflation after 2026. In 2019 prices this is **£70.9–82.2 bn** [54†L1-L4] . The government has endorsed a target in the lower part of this range (~£93.2 bn) to drive efficiencies [48†L159-L164] . Table 2 summarises these scenarios.

Table 2: HS2 Cost Scenarios (Phase 1)

Scenario	2019-price equivalent	Source
<i>Government target</i>	£70.9 bn – £82.2 bn	[3†L93-L100][54†L1-L4] (updated May 2026)
<i>HS2 Ltd ambition</i>	~£93.2 bn	[48†L159-L164] (DfT Ministerial Statement)
<i>Upper Estimate</i>	£102.7 bn	[18†L283-L290] (NAO summary)
<i>Previous baseline</i>	£38–55 bn	HS2 2013–2018 plans (pre-cancellation)

[52†embed_image] *Figure: HS2 Phase 1 expenditure to Mar 2026 and forecast by category. Civils are ~£31.2 bn spent (70% of total); Stations ~£4.8 bn [8†L800-L807] . Remaining works (notably systems and Rolling Stock, and future Civils) dominate the remaining budget. (Source: DfT 6-monthly report, May 2026.)*

The critical path runs through the main civil works and systems. Using the timeline above, the **long-lead items** are tunnel construction and station buildings (Curzon and OOC), followed by track/system installation. Any delay in civils pushes out tracklay. Euston station (not shown above) is a separate project with its own schedule; its completion (now expected in the early 2040s) will determine ultimate “full opening”, although through-running from the west is possible via Old Oak in late 2030s. Key schedule constraints: tunnel breakthroughs, shipping/installation of large spans, delivery of transformer stations, and software integration testing. Six months of float are estimated (see Gantt). Any significant slippage beyond those floats (e.g. if a TBM stalls) will threaten opening dates.

Stakeholder and Community Engagement

HS2 Ltd has engaged communities through a multi-channel approach [24†L87-L95] :

- **Forums and Consultations:** During design, HS2 ran regular statutory consultations and maintained Community and Environment Forums along the route [24†L87-L95] . These continue informally for local issues (landowner groups, environmental liaison committees).

Feedback on construction impacts is collected and addressed via the Code of Construction Practice.

- **Compensation and Support:** A dedicated Land and Property team handles compensation claims; a Community Engagement Panel (incl. local leaders) meets quarterly. Special arrangements (relocation grants, bonds) proposed by the Land Review will be deployed. HS2 Ltd reports on engagement metrics (number of complaints, average response time).
- **Local Authorities and MPs:** HS2 Ltd holds regular liaison meetings with affected councils and MPs, providing progress updates and addressing concerns (e.g. traffic management, local employment). The government also convenes project-specific task forces for hotspot areas (e.g. Birmingham), involving ministerial and local leadership.
- **Public Communication:** Ongoing publicity highlights benefits (journey times, jobs, regeneration). An external “Residents’ Commissioner” monitors household impacts and service quality in relocations. Clear channels for complaints (and a parliamentary Ombudsman pathway) are maintained.

No official “stop the project” campaign remains credible, given Government’s commitment [48†L208-L217] . Nevertheless, proactive engagement and maintaining a social license are critical – for example, HS2 Ltd must keep to promised community hires, apprenticeship quotas, and mitigate night-time noise. Any residual parliamentary scrutiny (e.g. by Transport Select Committee) will continue, so transparency of data (many reports are published on gov.uk [3†L780-L788]) is important.

Contingency and Next Steps

Given remaining uncertainties, plans include:

- **Contingency Planning:** A contingency fund (~10–15% of remaining cost) is held within the DfT budget. If cost or time exceed current estimates (e.g. due to unforeseen ground conditions or supply issues), drawdown must be approved by Treasury under defined authority. Critical “trigger points” (e.g. TBM completion, final Euston design) will be reassessed against plan.
- **Alternatives/Scope Adjustments:** HS2 Ltd will continue to review minor scope changes that could save cost or time (e.g. using proven UK-standard signalling instead of bespoke Austrian systems, or simplifying station finishes). Any such changes will be brought forward through the governance process (Shareholder Board approval).
- **Fallback Options:** In the very unlikely event of severe overruns, the Secretary of State could elect to pause works and seek Parliamentary review – though the cost of cancelation is judged comparable to completion [48†L208-L217] . A “pause and redesign” is the extreme last resort. Instead, the current strategy is to accelerate delivery efficiency under the reset.
- **Monitoring:** HS2’s progress will be reported to Parliament every 6 months. The NAO and Infrastructure Projects Authority will continue scrutiny of schedule/cost. If risk thresholds are crossed, remedial task forces (as convened by the 2025 reset) will be re-activated.

Data Gaps: Some details (e.g. precise cost breakdown for stations vs systems) have not been published; this plan assumes rough estimates from HS2 Ltd. The future shape of Northern connectivity (NPR) remains to be decided, so beyond guiding land use we have not planned those works here.

In conclusion, completing HS2 requires rigorous project control, clear decision-making (as instituted by the government reset 【48†L165-L174】) and disciplined execution of the steps above. If achieved, HS2 will deliver a transformational boost in UK rail capacity and journey times, as originally envisioned 【48†L225-L233】 【18†L293-L298】 . All strategic decisions and projections in this plan are based on the latest official reports 【48†L159-L164】 【18†L283-L290】 and planning documents, adjusted for current policy.
